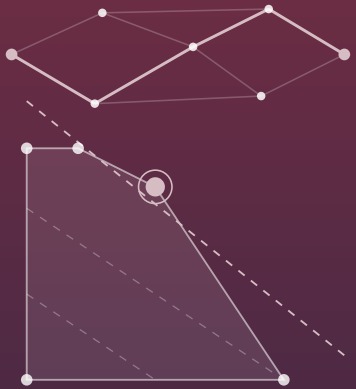
Operations Research

Lecture 2 · Semester 5



المدرسة الوطنية العليا لتكنولوجيا الأنظمة المستقلة

NHSAST

NATIONAL HIGHER SCHOOL OF AUTONOMOUS SYSTEMS TECHNOLOGY

Solving a Linear Program

The graphical method, the corner method, and all the possible cases

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

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Operations Research (O.R.) · Semester 5 · 2026–2027

Plan and goal

1. Three tools: draw a line, choose a side (arrow of coefficients), compute a corner
2. The graphical method on the drone workshop
3. Why the best plan is at a corner; the corner method; how to be sure
4. A minimisation example
5. All the cases: unique, many, unbounded, infeasible
6. Special situations: redundant constraint, degenerate corner, equality

Goal solve any LP with two variables and recognise every case.

Our example: the drone workshop (Lecture 1)

$$
\begin{aligned}
\max \quad & z = 40x_1 + 60x_2 && \text{(profit, thousand DA per week)} \\
\text{s.t.} \quad & x_1 + 2x_2 \leq 40 && \text{(assembly hours)} \\
& 3x_1 + 2x_2 \leq 60 && \text{(testing hours)} \\
& x_2 \leq 18 && \text{(cameras)} \\
& x_1, x_2 \geq 0
\end{aligned}
$$

- A plan (x_1, x_2) is a point of the plane.
- Each constraint keeps the points on one side of a line.
- The feasible set is the region where all constraints are satisfied.

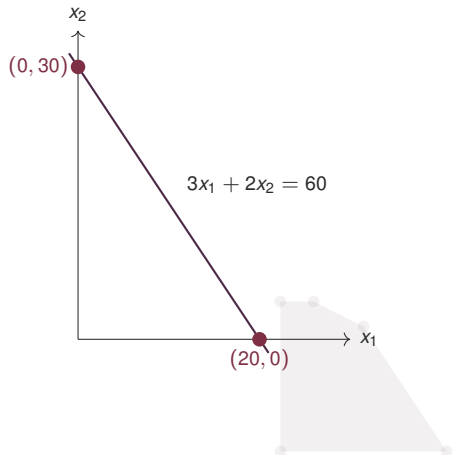
Tool 1: draw a line

The constraint $3x_1 + 2x_2 \leq 60$ is limited by the line $3x_1 + 2x_2 = 60$. Find two points:

- $x_1 = 0$: $2x_2 = 60$, $x_2 = 30$: point $(0, 30)$
- $x_2 = 0$: $3x_1 = 60$, $x_1 = 20$: point $(20, 0)$

Special lines:

- $x_2 = 18$: horizontal line
- $x_1 = 5$: vertical line
- $x_1 - x_2 = 0$: through the origin; use $(0, 0)$ and $(1, 1)$

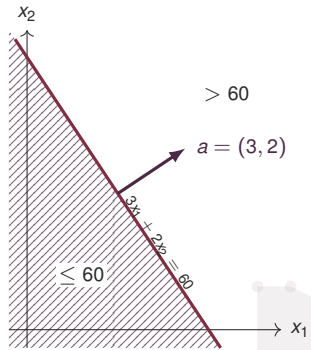


Tool 2: choose the correct side

The arrow of coefficients. For $3x_1 + 2x_2 \leq 60$, draw the arrow $a = (3, 2)$ from a point of the line.

- The arrow is perpendicular to the line.
- In its direction, $3x_1 + 2x_2$ grows: from $(0, 0)$ to $(3, 2)$ it goes from 0 to 13.
- So "$\leq$" keeps the side opposite to the arrow; "$\geq$" keeps the side of the arrow.

Check with a test point: $(0, 0)$ gives $0 \leq 60$, true. (Line through the origin, like $x_2 \leq 2x_1$: test $(1, 0)$.)



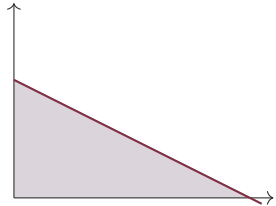
Tool 3 (Method 2.1): compute a corner

A corner is where two lines meet: solve the system of their two equations.

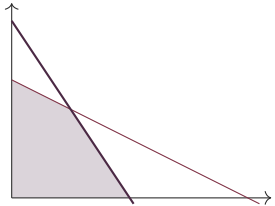
$$\begin{cases} x_1 + 2x_2 = 40 & (1) \\ 3x_1 + 2x_2 = 60 & (2) \end{cases}$$

- Elimination: (2) $-$ (1) gives $2x_1 = 20$, so $x_1 = 10$.
- Put $x_1 = 10$ in (1): $10 + 2x_2 = 40$, so $2x_2 = 30$ and $x_2 = 15$.
- Check in (2): $3 \times 10 + 2 \times 15 = 30 + 30 = 60$. Corner: $(10, 15)$.

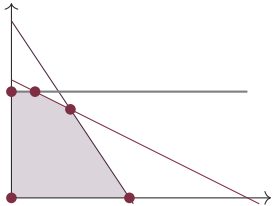
Building the feasible set, one constraint at a time



assembly
$x_1 + 2x_2 \leq 40$



+ testing
$3x_1 + 2x_2 \leq 60$



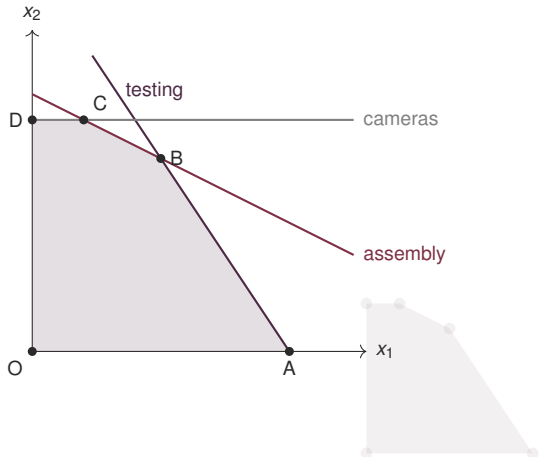
+ cameras
$x_2 \leq 18$

- Each constraint keeps a half-plane; the signs $x_1, x_2 \geq 0$ keep the first quarter.
- The feasible set is the part common to all the half-planes: a polygon.

Step 1: draw the feasible set

- Draw the three lines.
- $(0, 0)$ satisfies the three constraints: keep the side of $(0, 0)$ each time.
- With $x_1, x_2 \geq 0$, the feasible set is a polygon.

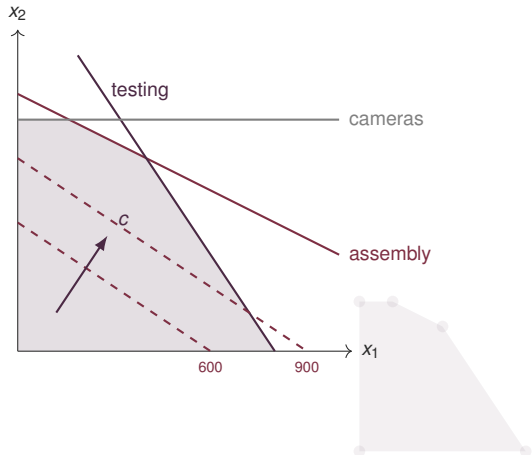
Corners: O $(0, 0)$, A $(20, 0)$, B $(10, 15)$, C $(4, 18)$, D $(0, 18)$.



Step 2: lines of equal profit

All plans with the same profit k lie on the line $40x_1 + 60x_2 = k$.

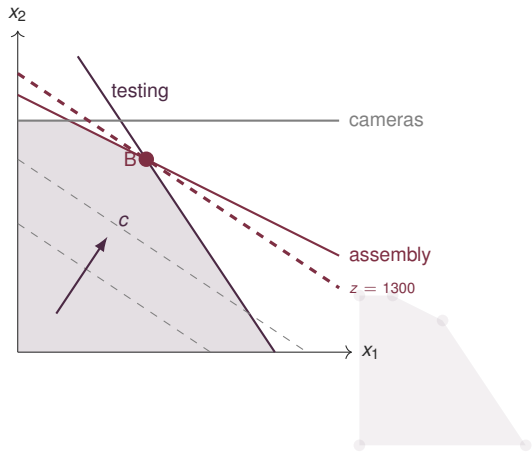
- $k = 600$: through $(15, 0)$ and $(0, 10)$
- $k = 900$: through $(22.5, 0)$ and $(0, 15)$
- These lines are parallel.
- The profit grows when the line moves up and to the right, in the direction of the arrow $c = (40, 60)$ of the coefficients of z (as in Tool 2).



Step 3: push the line as far as possible

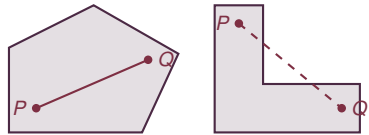
- Move the line in the direction of c while it still touches the feasible set.
- The last point touched is the corner B $= (10, 15)$.
- Optimal value: $z^* = 40 \times 10 + 60 \times 15 = 1300$ thousand DA.

Optimal plan: 10 drones A and 15 drones B per week. Whole numbers: by luck, it is also the best real plan (Lecture 1).



The feasible set has no holes and no dents

- If two plans P and Q are feasible, every plan on the segment between them is feasible.
- Why? Along the segment, each constraint expression changes regularly from its value at P to its value at Q; both are $\leq b$, so all values between are $\leq b$.
- A set with this property is called convex. The feasible set of an LP is always convex.



Key fact 2.1: the best plan is at a corner

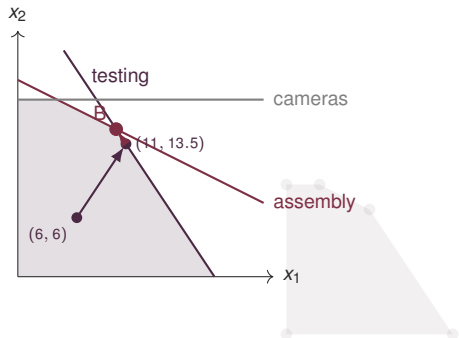
Key fact 2.1

If a linear program has an optimal solution, then at least one corner of the feasible set is optimal.

Sliding to a corner:

1. Start from a feasible plan: $(6, 6)$, $z = 600$.
2. Move in the direction of c until the border: $(11, 13.5)$, $z = 1250$.
3. Move along the border, where z does not decrease, to a corner: B, $z = 1300$.

From any plan we reach a corner at least as good.



Why the value changes regularly along an edge

Edge from C $(4, 18)$ to B $(10, 15)$: to go from C to B we add 6 to x_1 and -3 to x_2.

- Its points: $(4 + 6t,\ 18 - 3t)$, t from 0 (at C) to 1 (at B).
- Their profit: $40(4 + 6t) + 60(18 - 3t) = 160 + 240t + 1080 - 180t = 1240 + 60t$.
- A straight-line function of t: on the edge, the best value is at an end, here B.

t	0	0.25	0.5	0.75	1
Point	C $(4, 18)$	$(5.5, 17.25)$	$(7, 16.5)$	$(8.5, 15.75)$	B $(10, 15)$
z	1240	1255	1270	1285	1300

Method 2.2: the corner method

1. Draw the feasible set and find all its corners (Method 2.1).
2. Compute z at each corner.
3. Choose the best corner.

Corner	(x_1, x_2)	$z = 40x_1 + 60x_2$	
O	$(0, 0)$	0	
A	$(20, 0)$	800	
B	$(10, 15)$	1300	best
C	$(4, 18)$	1240	
D	$(0, 18)$	1080	

Warning: if the feasible set is unbounded, first check that the optimum exists (Cases 3 and 4).

Reading the solution

At the optimal plan $\mathrm{B} = (10, 15)$:

Constraint	Used	Available	Unused (slack)	
Assembly	$10 + 30 = 40$	40	0	binding
Testing	$30 + 30 = 60$	60	0	binding
Cameras	15	18	3	not binding

- A constraint is binding (active) when it is satisfied with equality: its line passes through the optimal point.
- Binding constraints are the bottlenecks: to earn more, we need more assembly or testing hours. More cameras would not help.

How can we be sure that 1300 is the best?

Idea: multiply constraints by positive numbers and add them (the inequality keeps its direction).

- **Try 1:** $40 \times (x_1 + 2x_2 \leq 40)$: $40x_1 + 80x_2 \leq 1600$. Since $x_2 \geq 0$:

$$40x_1 + 60x_2 \leq 40x_1 + 80x_2 \leq 1600.$$

  No plan beats 1600, but 1600 is not reached.

- **Try 2:** $25 \times (x_1 + 2x_2 \leq 40) \; + \; 5 \times (3x_1 + 2x_2 \leq 60)$:

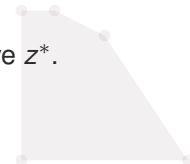
$$25x_1 + 50x_2 + 15x_1 + 10x_2 \leq 1000 + 300, \quad \text{that is} \quad 40x_1 + 60x_2 \leq 1300.$$

  No plan beats 1300, and B gives 1300: B is optimal.

How to find 25 and 5? Ask for $40x_1 + 60x_2$ on the left: $y_1 + 3y_2 = 40$ and $2y_1 + 2y_2 = 60$, so $y_2 = 5$, $y_1 = 25$.

Method 2.3: the graphical method (summary)

1. Draw each constraint line and keep the correct side (test point).
2. Shade the feasible set. If it is empty: the problem is infeasible. Stop.
3. Draw one line of equal value $z = k$ and the arrow c.
4. Push the line in the direction of c for a max, or in the opposite direction for a min, as far as possible while it touches the feasible set.
5. If it can be pushed forever: the problem is unbounded. Stop.
6. Otherwise, read the last corner (or edge) touched, compute it exactly, give z^*.



A minimisation example: spare-part boxes

A maintenance team buys boxes of spare parts for its robots.

	Box A	Box B	Needed (at least)
Motors per box	2	1	8
Sensors per box	1	2	7
Price (thousand DA per box)	30	40	

x_1, x_2 = numbers of boxes A and B bought:

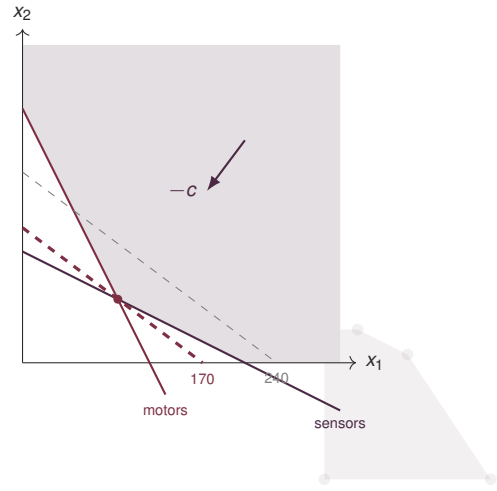
$$\min\ z = 30x_1 + 40x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \geq 8, \quad x_1 + 2x_2 \geq 7, \quad x_1, x_2 \geq 0.$$

Your turn · 3 min draw the feasible set and solve.

Spare-part boxes: the solution

- Test $(0,0)$: $0 \geq 8$ is false: keep the sides away from the origin.
- Corner: $2x_1 + x_2 = 8$ and $x_1 + 2x_2 = 7$. From the first, $x_2 = 8 - 2x_1$; then $x_1 + 16 - 4x_1 = 7$, so $x_1 = 3$, $x_2 = 2$.
- Corners: $(0,8)$: 320; $(3,2)$: 170; $(7,0)$: 210.
- Push the line towards the origin: the last point is $(3,2)$.

Buy 3 boxes A and 2 boxes B, for 170 thousand DA.



All the possible cases

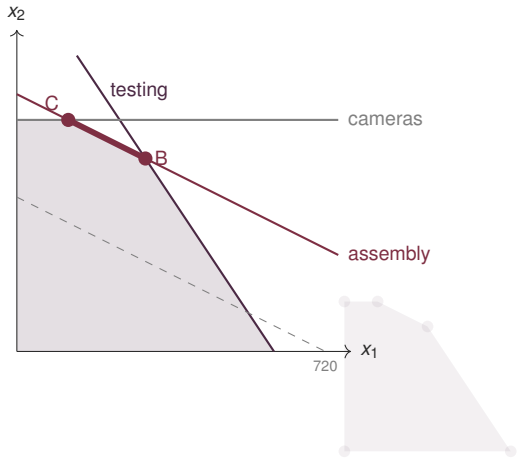
Case	What we see	Answer
1. Unique optimum	the line touches one corner last	one optimal plan
2. Many optima	the line touches an edge last	infinitely many optimal plans
3. Unbounded set, optimum exists	the set goes to infinity, but the line stops	an optimal plan exists
4. Unbounded problem	the line can be pushed forever	no optimal plan
5. Infeasible problem	the feasible set is empty	no feasible plan

After the five cases: three special situations (redundant constraint, degenerate corner, equality constraint).

Case 2: infinitely many optimal solutions

If the profit of a drone A is 30 thousand DA: $z = 30x_1 + 60x_2$.

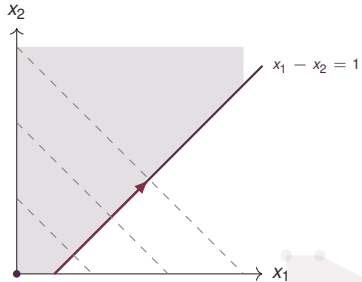
- The lines $30x_1 + 60x_2 = k$ are parallel to the assembly line $x_1 + 2x_2 = 40$.
- Corners: B gives $300 + 900 = 1200$, C gives $120 + 1080 = 1200$: same best value.
- All points of the edge from C to B are optimal: $(4 + 6t,\ 18 - 3t)$, $0 \le t \le 1$.



Cases 3 and 4: an unbounded feasible set

Feasible set: $x_1 - x_2 \leq 1,\ x_1, x_2 \geq 0$ (it goes to infinity).

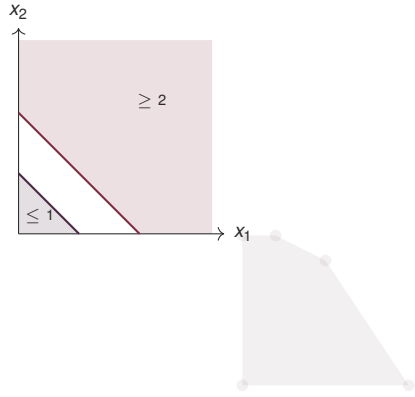
- **Case 3**: min $x_1 + x_2$. Pushing towards the origin, the line stops at $(0, 0)$: optimum $(0, 0)$, $z^* = 0$.
- **Case 4**: max $x_1 + x_2$. The line can be pushed forever. Check: the points $(1 + t,\ t)$ are feasible for every $t \geq 0$, and $z = 1 + 2t$ becomes as large as we want. No optimal plan.



Case 5: no feasible solution

$\max\ x_1 + x_2$ s.t. $x_1 + x_2 \leq 1,\ x_1 + x_2 \geq 2,$ $x_1, x_2 \geq 0.$

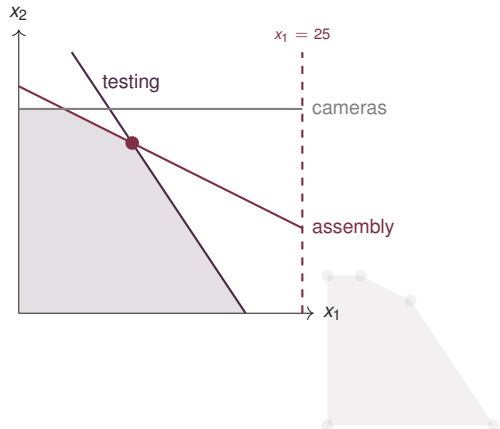
- The two sides do not meet: the feasible set is empty.
- With numbers: we would need $2 \leq x_1 + x_2 \leq 1$, impossible.
- In practice: an error in the data or in the model, or a demand that is too high (Lecture 1, transport).



Special situation A: a redundant constraint

Add to the drone workshop: “at most 25 drones A per week”: $x_1 \leq 25$.

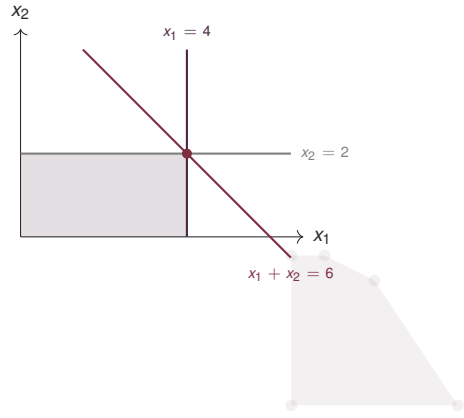
- The line $x_1 = 25$ is outside the polygon: the testing constraint already gives $x_1 \leq 20$.
- The feasible set does not change, and the optimum is still $\mathrm{B} = (10, 15)$.
- Such a constraint is redundant: we can remove it.



Special situation B: a degenerate corner

$\max\ 2x_1 + x_2$ s.t. $x_1 \le 4,\ x_2 \le 2,$
$x_1 + x_2 \le 6,\ x \ge 0.$

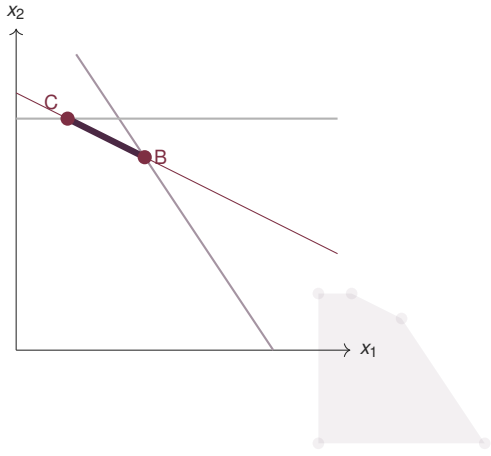
- Three lines pass through the corner $(4, 2)$: $4 \le 4$, $2 \le 2$, $4 + 2 = 6$.
- Such a corner is degenerate.
- Graphically, no problem: corners $(0, 0)$: 0, $(4, 0)$: 8, $(4, 2)$: 10, $(0, 2)$: 2.
- It matters for the simplex method (later).



Special situation C: an equality constraint

Drone workshop with “all 40 assembly hours must be used”: $x_1 + 2x_2 = 40$.

- The feasible set is no longer a polygon: it is the segment of the line $x_1 + 2x_2 = 40$ inside the other constraints, from C $(4, 18)$ to B $(10, 15)$.
- Its corners are its two ends: C gives 1240, B gives 1300.
- Optimum: B $= (10, 15)$.



Method 2.4: find the corners without drawing

A corner = a point where two constraint lines meet, and which satisfies all the constraints.

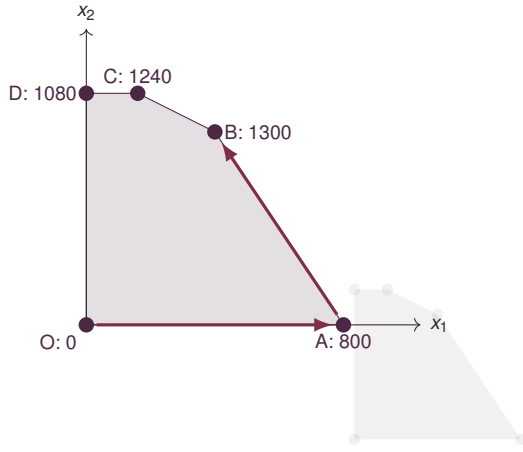
Two lines		Point	Feasible?	z
assembly	testing	$(10, 15)$	yes	1300
assembly	cameras	$(4, 18)$	yes	1240
assembly	$x_1 = 0$	$(0, 20)$	no: $20 > 18$ cameras	
assembly	$x_2 = 0$	$(40, 0)$	no: testing $120 > 60$	
testing	cameras	$(8, 18)$	no: assembly $44 > 40$	
testing	$x_1 = 0$	$(0, 30)$	no	
testing	$x_2 = 0$	$(20, 0)$	yes	800
cameras	$x_1 = 0$	$(0, 18)$	yes	1080
cameras	$x_2 = 0$	parallel lines	—	
$x_1 = 0$	$x_2 = 0$	$(0, 0)$	yes	0

Walking from corner to corner

1. Start at a corner: O, $z = 0$.
2. Move along an edge to a neighbouring corner with a better value: A (800), then B (1300).
3. At B, the neighbours A (800) and C (1240) are worse: stop. B is optimal.

Another path also works: O $\rightarrow$ D (1080) $\rightarrow$ C (1240) $\rightarrow$ B.

This is the idea of the simplex method (next lectures).



What if a profit changes?

Profit of a drone A $= c_1$ (instead of 40). Values at the three corners near B:

$$z(\mathrm{A}) = 20c_1, \qquad z(\mathrm{B}) = 10c_1 + 900, \qquad z(\mathrm{C}) = 4c_1 + 1080.$$

- B better than C: $10c_1 + 900 \geq 4c_1 + 1080 \iff 6c_1 \geq 180 \iff c_1 \geq 30.$
- B better than A: $10c_1 + 900 \geq 20c_1 \iff 900 \geq 10c_1 \iff c_1 \leq 90.$

The plan $(10, 15)$ stays optimal for $30 \leq c_1 \leq 90$. At $c_1 = 30$ or $c_1 = 90$: many optima (Case 2).

How to recognise each case

If you see…	then
the feasible set is empty	Case 5: infeasible
the line can be pushed forever (points $x(t)$ with $z \to \infty$)	Case 4: unbounded
the last contact is one corner	Case 1: unique optimum
two neighbouring corners give the same best value	Case 2: infinitely many optima
a line outside the set, or touching only one corner	redundant constraint
three or more lines through one corner	degenerate corner

Summary and next lecture

- Tools: draw a line (two points), choose a side (test point), compute a corner (solve two equations).
- Graphical method: push the line of equal value in the direction of c (max) or $-c$ (min).
- Key fact 2.1: if an optimum exists, a corner is optimal $\rightarrow$ corner method; combining constraints proves optimality.
- Five cases: unique, many, unbounded set with optimum, unbounded problem, infeasible.
- Special situations: redundant constraint, degenerate corner, equality.
- TD 2: solve the models of TD 1, and all the cases. Next: the simplex method.